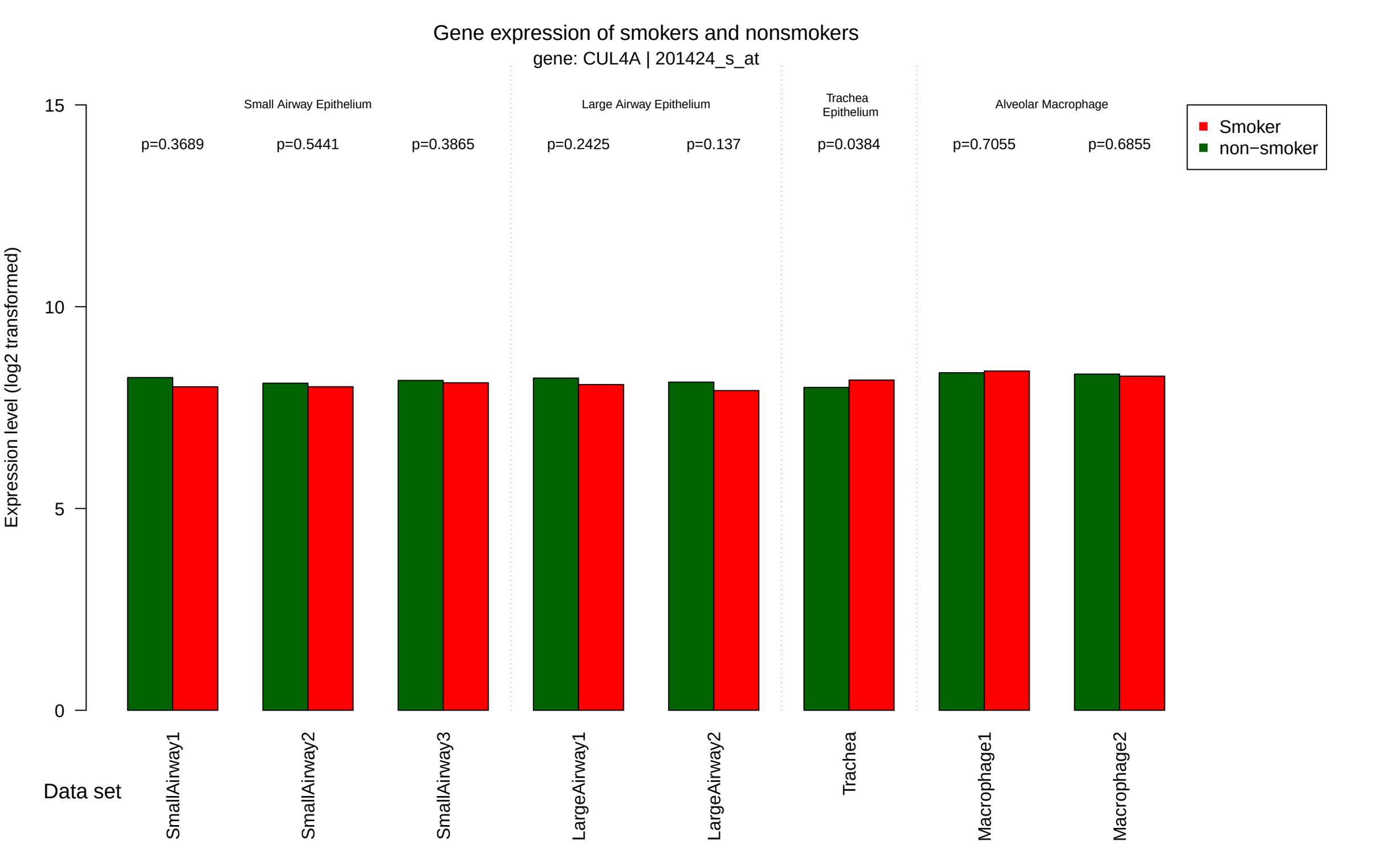
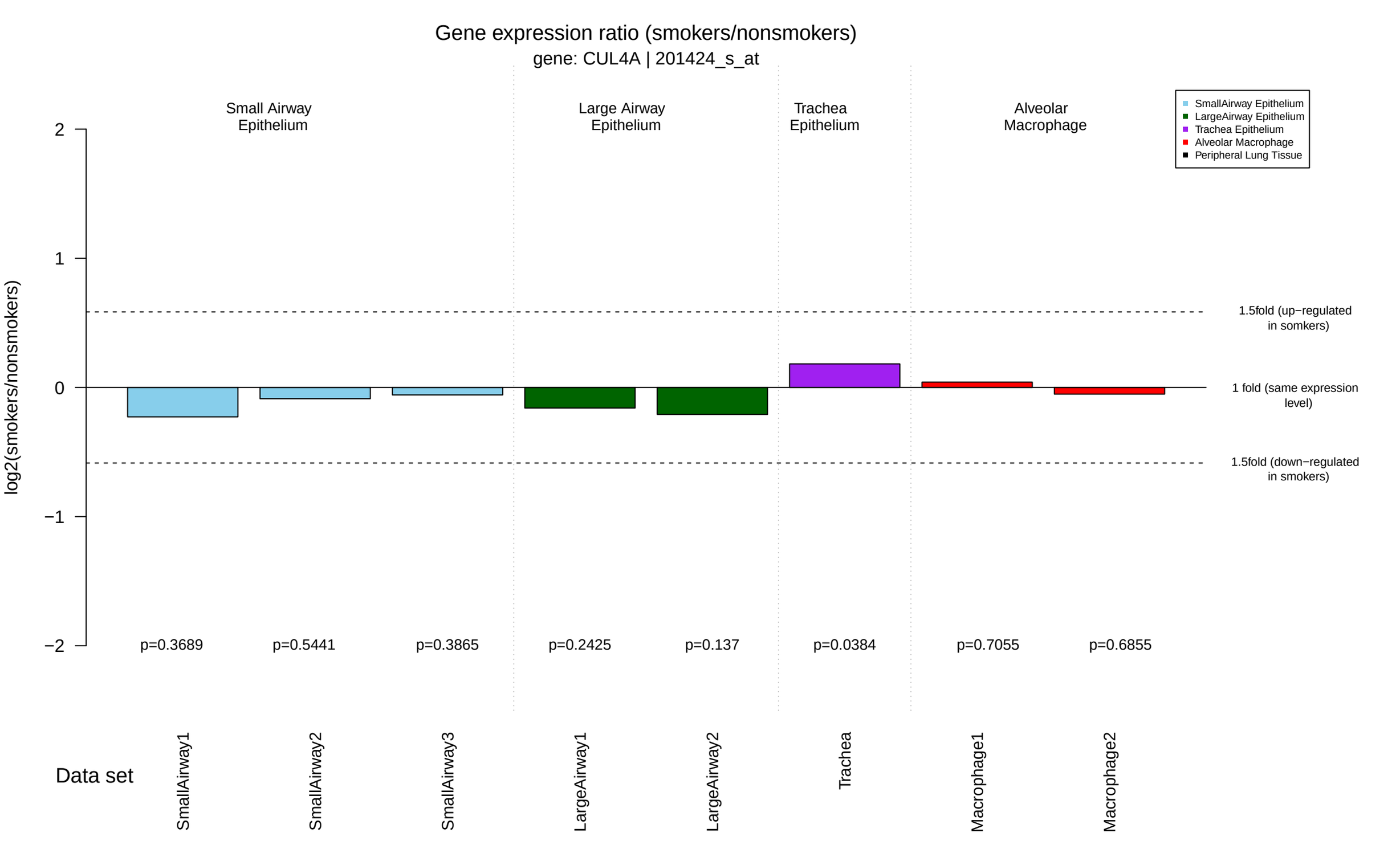
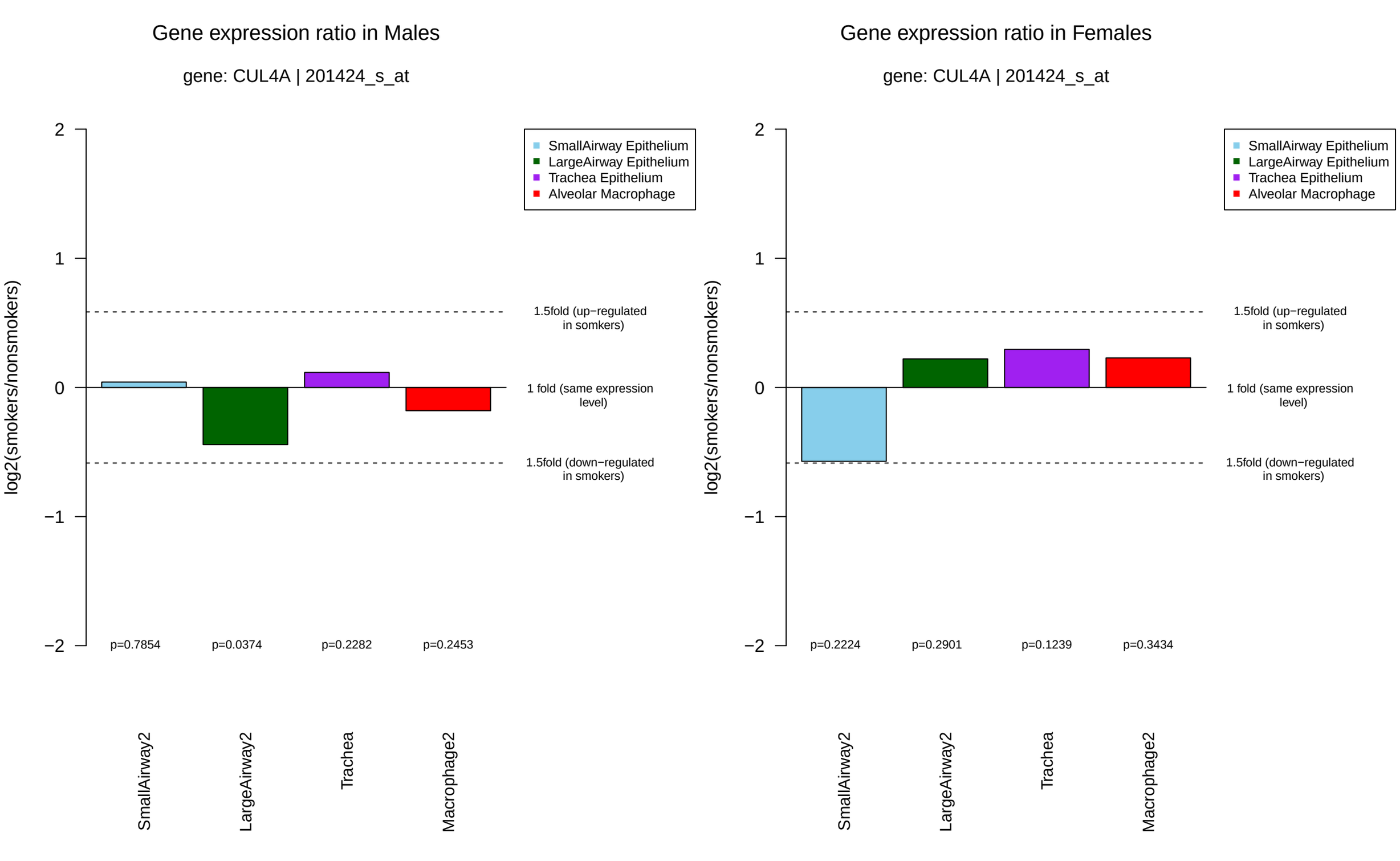
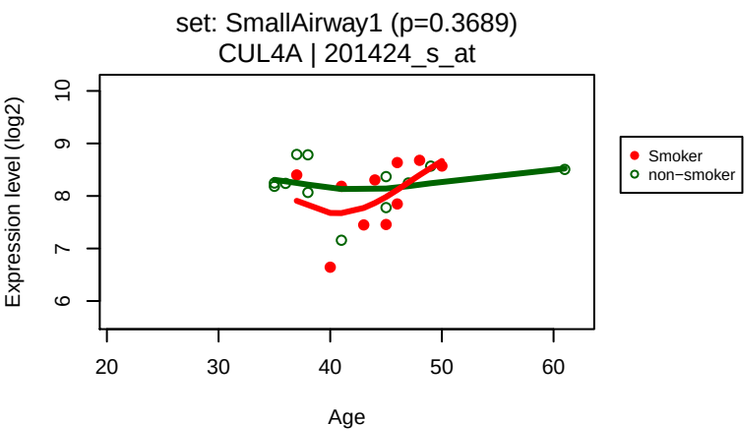
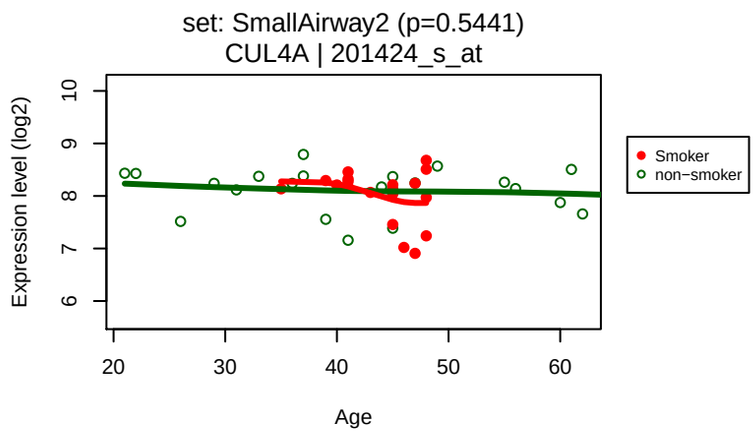
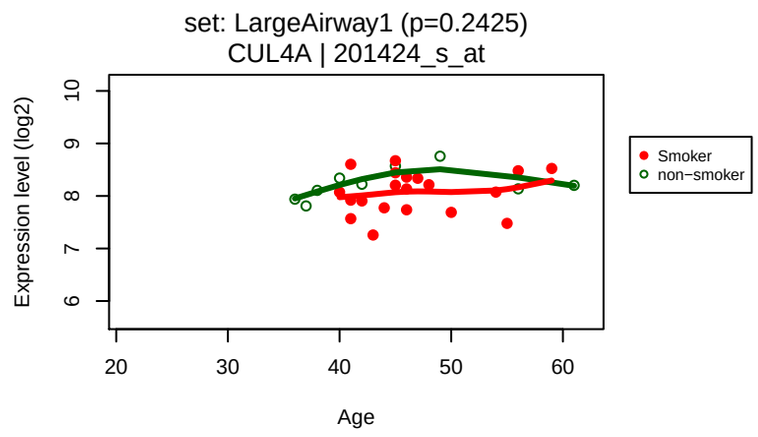

Gene expression of smokers and nonsmokers
gene: CUL4A | 201424_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3689
p=0.5441
p=0.3865
p=0.2425
p=0.137
p=0.0384
p=0.7055
p=0.6855
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

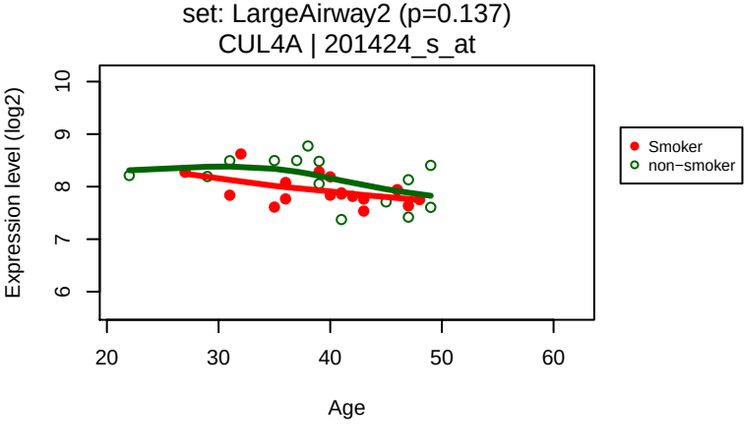

Gene expression ratio (smokers/nonsmokers)
gene: CUL4A | 201424_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3689
p=0.5441
p=0.3865
p=0.2425
p=0.137
p=0.0384
p=0.7055
p=0.6855
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

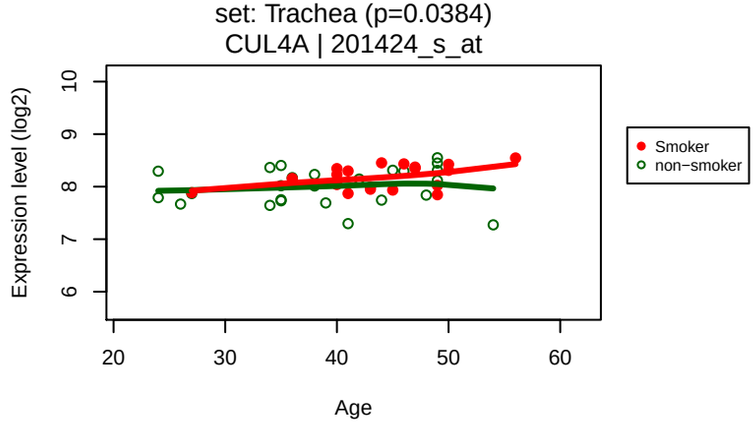

Gene expression ratio in Males
gene: CUL4A | 201424_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7854
p=0.0374
p=0.2282
p=0.2453
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CUL4A | 201424_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2224
p=0.2901
p=0.1239
p=0.3434
SmallAirway2
LargeAirway2
Trachea
Macrophage2

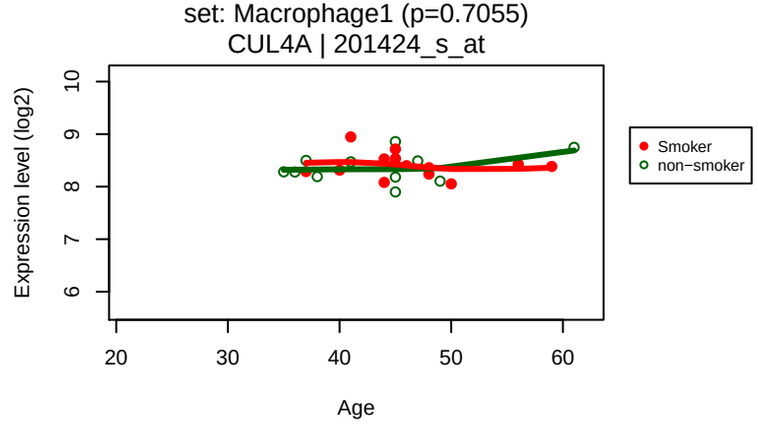

set: SmallAirway1 (p=0.3689)
CUL4A | 201424_s_at
Expression level (log2)
Age
Smoker
non−smoker

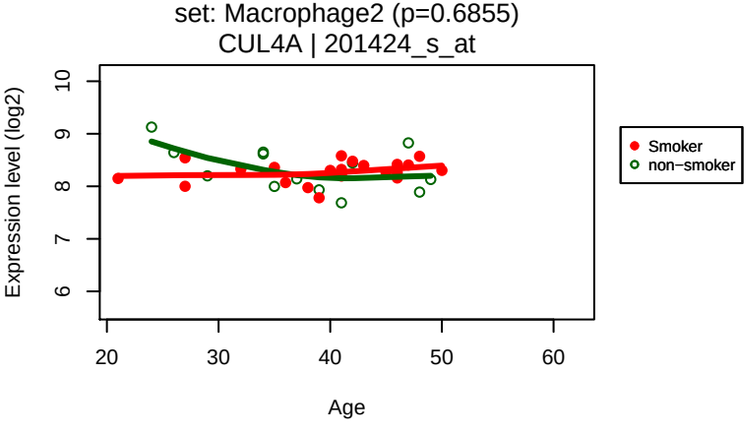

set: SmallAirway2 (p=0.5441)
CUL4A | 201424_s_at
Expression level (log2)
Age
Smoker
non−smoker

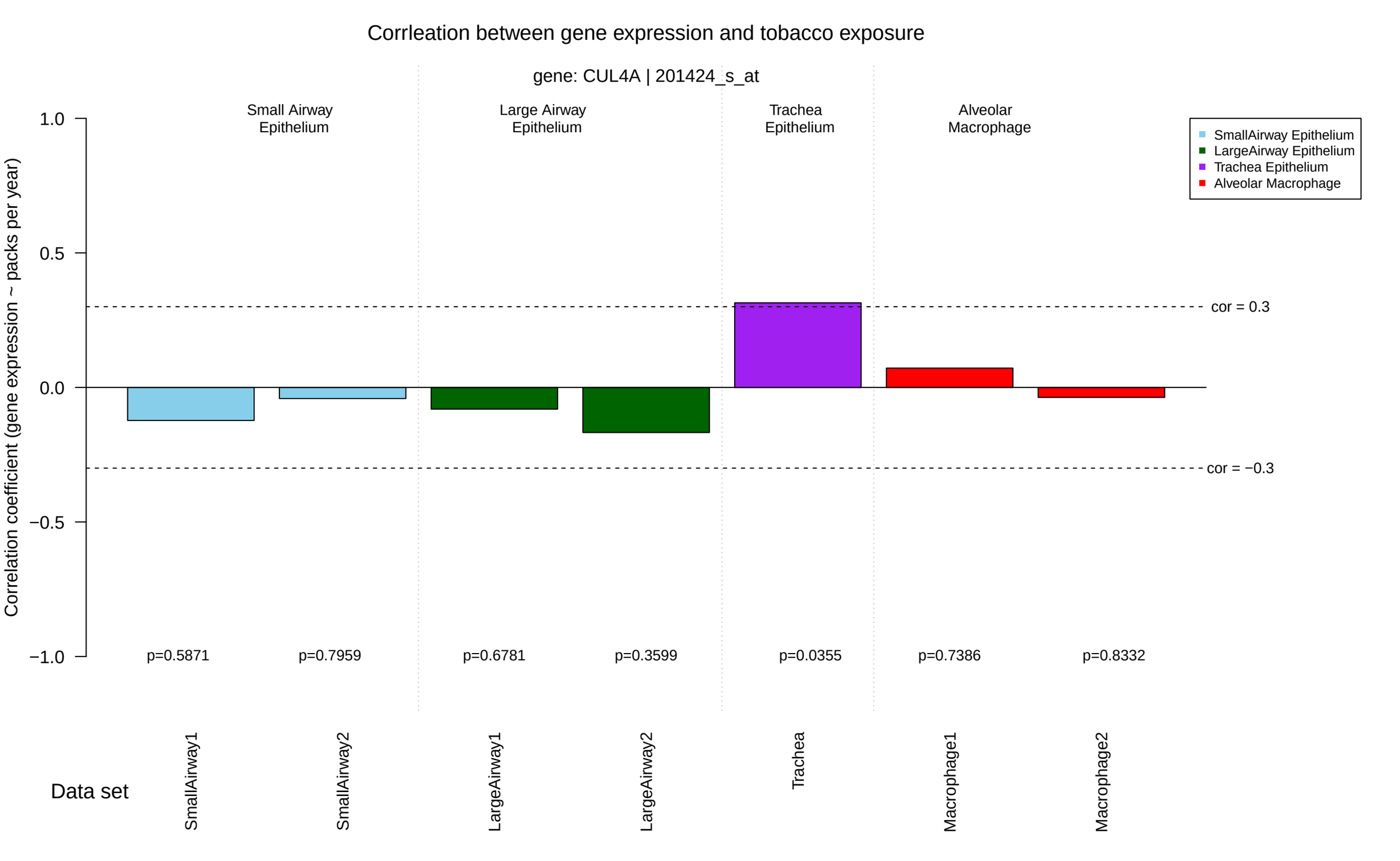

set: LargeAirway1 (p=0.2425)
CUL4A | 201424_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.137)
CUL4A | 201424_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0384)
CUL4A | 201424_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7055)
CUL4A | 201424_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6855)
CUL4A | 201424_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CUL4A | 201424_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5871
p=0.7959
p=0.6781
p=0.3599
p=0.0355
p=0.7386
p=0.8332
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2